

AI-Based Vision Assistance System

For Visually Impaired Using Computer Vision

Project Category: Artificial Intelligence and Computer Vision

Generated on: March 04, 2026

System Version: 1.0.0

SYNOPSIS

Visual impairment significantly limits a person's ability to understand their surroundings and read printed information independently. With recent advancements in artificial intelligence and computer vision, it has become possible to analyze visual data and convert it into meaningful audio feedback.

This project focuses on developing a software-based vision assistance system that helps visually impaired users by identifying surrounding objects and reading printed text using a laptop webcam. The system aims to provide real-time audio assistance without the need for specialized hardware.

Objectives:

- To develop a real-time object detection system using computer vision
- To provide audio feedback for detected objects
- To implement text reading functionality using OCR
- To create a cost-effective, software-based assistance system
- To demonstrate the system using a laptop without external hardware

Project category:

This project belongs to the Artificial Intelligence and Computer Vision category. It is a software-based application that uses machine learning, image processing, and text-to-speech techniques to assist visually impaired users. The system analyzes real-time visual input using computer vision algorithms and provides audio feedback without requiring any specialized hardware.

TOOLS AND PLATFORMS

Tools:

- Visual Studio Code / PyCharm (Code Editor)
- Anaconda Distribution for Python
- OpenCV Library
- TensorFlow / PyTorch
- Tesseract OCR / EasyOCR
- Text-to-Speech Library (pyttsx3 / gTTS)

Platform:

- Web Camera (Laptop in-built)
- Operating System: Windows / Linux / macOS

REQUIREMENTS

Hardware Requirements:

- Processor: Intel Core i5 or higher
- RAM: 4 GB or more
- Storage: Minimum 100 GB HDD / SSD
- Webcam and Speaker
- Internet connection (for setup and testing)

Client Requirements (User Device):

- Compatible laptop or desktop system
- In-built webcam
- Speaker for audio output
- Latest version of operating system (Windows / Linux / macOS)

Software Requirements:

- Programming Language: Python
- Computer Vision Library: OpenCV
- AI / Machine Learning Framework: TensorFlow or PyTorch
- OCR Tool: Tesseract OCR
- Text-to-Speech Engine: pyttsx3 / gTTS
- Development Environment: Visual Studio Code / PyCharm
- Platform: Anaconda (Python environment)

SETUP & EXECUTION

Initial Configuration:

1. Extract the project source folder.
2. Open terminal in the project directory.
3. Run './start.sh' to initialize and launch system.

Running the Application:

After execution, open Chrome/Edge and browse to <http://localhost:5173>.

The system will initialize the camera and start the AI detection loop automatically.